Project Initialization and Planning Phase

Date	03 July 2026
Team ID	
Project Title	Credit Card Approval Prediction
Maximum Marks	5 Marks

Project Proposal (Proposed Solution) report

The proposed project aims to automate the credit card approval process using Machine Learning, improving prediction accuracy, reducing manual effort, and enabling faster approval decisions. The system analyzes applicant information such as income, employment status, family status, housing type to predict whether a credit card application should be approved or rejected. The project also includes a Flask-based web application for easy user interaction , real-time prediction and local execution.

Project Overview	
Objective	The primary objective is to develop an AI-powered Credit Card Approval Prediction System that accurately predicts whether a credit card application should be approved or rejected using Machine Learning algorithms, thereby reducing processing time and improving decision accuracy.
Scope	The project covers data collection, preprocessing, visualization, feature engineering, model training, model evaluation, and deployment through a Flask web application. It provides fast and reliable credit card approval predictions for banking institutions.
Problem Statement	
Description	The existing credit card approval process relies heavily on manual verification, making it slow, time-consuming, and prone to human errors. Applicants often experience delays and uncertainty regarding their application status.
Impact	Implementing an AI-based prediction system will reduce approval time, improve prediction accuracy, minimize manual work, increase operational efficiency, and enhance customer satisfaction.
Proposed Solution	
Approach	Develop a Machine Learning model using applicant demographic and financial information to predict credit card approval. The best-performing model is integrated into a Flask web application for real-time prediction and deployed using IBM Watson Machine Learning.

Key Features	• Machine Learning-based approval prediction • Real-time approval or rejection prediction • Data preprocessing and feature engineering • Comparison of multiple ML algorithms • Flask-based web application • User-friendly interface • High prediction accuracy • Prediction History • Dashboard with Pie and Bar Charts • CSV Download

Resource Requirements

Resource Type	**Description**	**Specification/Allocation**
Hardware		
Computing Resources	CPU	Intel i3/i5 or equivalent
Memory	RAM	4 GB minimum (8 GB recommended)
Storage	Disk space	20 GB Free space
Software		
Frameworks	Backend	Flask
Libraries	ML Libraries	scikit-learn, pandas, numpy, matplotlib, joblib
Development Environment	IDE	Jupyter Notebook, VSCode
Data		
Dataset	Source, size, format	Kaggle Credit Card Dataset, CSV format